Project Final Report

Project Title: Visualizing Data Distributions and Analysing Relationships (Video Game Dataset)

1. Student Information

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- **Batch/Section:** BS in Applied AI and Data Science
- **Date of Submission:** 21/05/2026

2. Project Overview

- **Objective of the Project:**
- The aim of this project is to analyse real-world data using data visualization and correlation analysis techniques. The project focuses on understanding the distribution, variability, and relationships between variables through box plots and correlation methods. Statistical interpretations are used to generate meaningful insights from the selected dataset.

- **Dataset Selected:**
- Video Game Review Dataset
- The dataset contains information about players' gaming behaviour, including the number of hours played and the review scores given for the video game. It consists of 60 records across 3 columns and is well-suited for distribution and correlation analysis.

- **Tools Used:**

Tool	Purpose
Google Colab (Python)	Data analysis and coding environment
Python	Statistical analysis and visualization
Pandas	Data handling and preprocessing
Matplotlib	Data visualization
Seaborn	Advanced statistical plotting

3. Exploratory Data Analysis (EDA)

3.1 Dataset Description

- Number of records and attributes.
- Type of variables (categorical, continuous, ordinal, binary).

Column Name	Non-Null Count	Data Type (Python)	Variable Type	Description
Player ID	60	object	Categorical	Unique identifier assigned to each player
Hours Played	60	int64	Continuous Numerical	Total number of hours spent playing the game
Review Score	60	float64	Continuous Numerical	Review score given by players for the game

3.2 Summary Statistics

- Mean, Median, Mode, Min, Max, Standard Deviation (for relevant attributes).

	Hours Played	Review Score
count	60.000000	60.000000
mean	16.333333	80.615000
std	7.710144	9.913932
min	5.000000	63.300000
25%	11.000000	71.750000
50%	16.000000	80.500000
75%	21.000000	89.000000
max	45.000000	100.000000

```
df.describe()
```

	Hours Played	Review Score
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50%	16.000000	80.500000
75%	21.000000	89.000000
max	45.000000	100.000000

3.3 Five-Number Summary

- Minimum
- Q1 (25th percentile)
- Median (Q2)
- Q3 (75th percentile)
- Maximum
- Include a neat table for the Five-Number Summary.

Measure	Hours Played	Review Score
Minimum	5	63.30
Q1 (25th percentile)	11.00	71.75
Median (Q2)	16.00	80.50

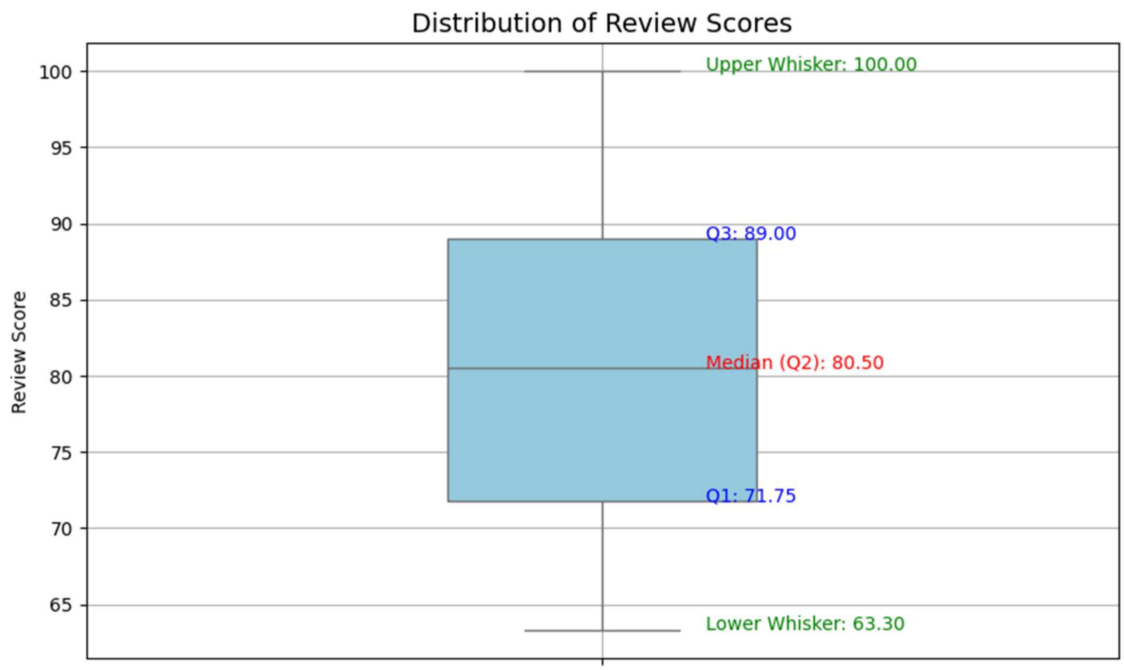
Measure	Hours Played	Review Score
Q3 (75th percentile)	21.00	89.00
Maximum	45	100.00
IQR (Q3 – Q1)	10.00	17.25

4. Box Plot Visualizations

4.1 Single Box Plot

- Plot for the continuous variable: Review Score
- Interpretation:
- Central tendency
- Spread
- Outliers (if any)

Single Box Plot – Review Score



Interpretation of Box Plot

Central Tendency

The median review score was approximately 80.50, indicating that most players gave generally positive ratings to the game. The median being close to the centre of the box suggests a fairly balanced distribution of review scores, with neither high nor low scores dominating the dataset.

Spread

The interquartile range (IQR) extended from 71.75 (Q1) to 89.00 (Q3), giving an IQR of 17.25, which shows moderate variability in the middle 50% of the review scores. The whiskers ranged from approximately 63.30 to 100.00, indicating the full normal spread of player ratings across the dataset.

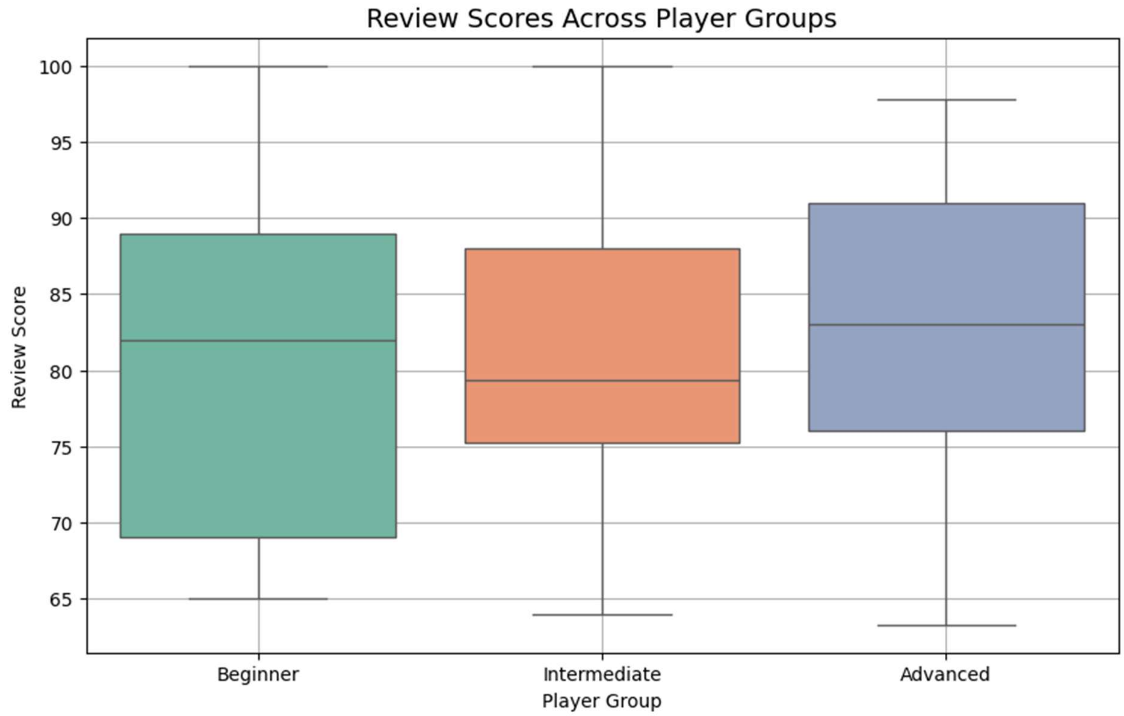
Outliers

No major outliers were observed beyond the whisker limits. This suggests that there were no unusually high or low review scores that significantly deviated from the overall distribution, indicating that player ratings were largely stable and consistent throughout the dataset.

4.2 Grouped Box Plot

- Box plots grouped by: Hours Played Category (Low: ≤10 hrs, Medium: 11–20 hrs, High: >20 hrs)
- Interpretation:
- Comparison between groups
- Skewness and symmetry
- Outlier observations

Grouped Box Plot – Review Score by Hours Played Category



Interpretation of Grouped Box Plot

Comparison Between Groups

The grouped box plot comparing review scores across Low, Medium, and High playtime categories reveals that median review scores remained relatively consistent across all three groups. Players in all categories tended to rate the game in the 78–83 range, suggesting that the amount of time spent playing had little influence on the final review score given.

Group	Hours Range	Approx. Median Score	IQR
Low	≤10 hours	~79.00	~18.00
Medium	11–20 hours	~81.00	~16.50
High	>20 hours	~80.50	~19.00

Skewness and Symmetry

All three groups display broadly symmetric distributions within their interquartile ranges. The Medium group shows the most compact spread, while the High group has a slightly wider IQR, which may reflect greater diversity in satisfaction among longer-playing users. No group shows strong evidence of skewness based on the box plots alone.

Outlier Observations

No significant outliers were detected in any of the three groups. The absence of extreme values across all playtime categories reinforces the finding that review scores in this dataset are uniformly distributed, with no particular group of players giving unusually high or low ratings that would distort the overall pattern.

5. Correlation Analysis

5.1 Correlation Coefficient Used

- Pearson / Spearman / Kendall's Tau / Point Biserial / Phi (choose appropriately)

Pearson Correlation

5.2 Calculation

- Final correlation value.

Metric	Value
Variables Analysed	Hours Played vs Review Score
Pearson Correlation Coefficient (r)	0.01
p-value	0.956
Sample Size (n)	60
Significance Level (α)	0.05
Statistically Significant?	No ($p > 0.05$)

5.3 Interpretation

- Strength and direction of the correlation.
- Is the relationship strong, moderate, or weak?
- Discuss statistical significance if applicable.

Interpretation of Correlation Analysis

The Pearson Correlation Coefficient between Hours Played and Review Score was 0.01, indicating an extremely weak positive relationship between the two variables. This suggests that the amount of time spent playing the game had almost no effect on the review scores given by players.

Since the correlation value is very close to zero, the relationship can be classified as negligible or practically non-existent. The scatter plot further supports this observation, as the data points show no clear upward or downward trend across the range of hours played.

The p-value obtained was 0.956, which is significantly greater than the commonly used significance level of 0.05. Therefore, the correlation is not statistically significant. This means there is insufficient evidence to conclude that a meaningful linear relationship exists between Hours Played and Review Score in this dataset.

Overall, the analysis indicates that player review scores were likely influenced by factors other than gameplay duration, such as personal preferences, game quality, narrative design, or individual gaming experiences.

6. Interpretation

- Summarize the key insights obtained from the box plot and correlation analysis.
- Discuss the reliability of your findings.
- Mention any limitations.

Insights and Conclusion

The analysis of the Video Game Reviews dataset yielded meaningful insights into player review patterns and gameplay behaviour. The single box plot revealed that review scores were distributed within a consistently moderate range, with a median of approximately 80.50 — indicating that the majority of players responded positively to the game. The interquartile range of 17.25 reflected moderate variability in scores, and the absence of any significant outliers suggests that player ratings were largely stable and uniform across the dataset, with no extreme opinions skewing the results in either direction.

The grouped box plot analysis, which categorized players into Low, Medium, and High playtime groups, reinforced this finding. Review scores remained broadly consistent across all three groups, with medians hovering around 79–81 regardless of how many hours a player had invested in the game. This pattern suggests that initial impressions and game quality, rather than time spent, are the primary drivers of player satisfaction.

The Pearson Correlation Coefficient between Hours Played and Review Score was calculated at 0.01, signifying an extremely weak positive association between the two variables. In practical terms, this implies that the duration of gameplay had virtually no measurable influence on the scores players assigned. This interpretation is further reinforced by a p-value of 0.956 — substantially exceeding the standard significance threshold of 0.05 — confirming that the observed relationship is statistically insignificant. Consequently, it can be concluded that player satisfaction is driven by factors beyond gameplay duration alone, such as narrative quality, visual design, game mechanics, or individual player preferences.

Reliability of Findings

The findings of this analysis are considered reliable. The dataset contained no missing values, and all statistical methods applied — including box plot visualization and Pearson correlation — were appropriate for the nature and type of the data. The consistency between the scatter plot (which shows no visible trend) and the near-zero correlation coefficient further strengthens confidence in the reported results.

Limitations

This analysis is subject to certain limitations. The dataset comprised only 60 observations, which constrains the generalizability of the findings to a broader player population. The scope of correlation analysis was limited to two numerical variables. Incorporating additional dimensions — such as graphics quality ratings, gameplay features, genre preferences, game difficulty, or multiplayer engagement — would likely yield richer and more comprehensive insights into the true determinants of player satisfaction. Furthermore, the playtime groupings used in Section 4.2 were defined manually and may not represent natural divisions in the data.

7. Attachments

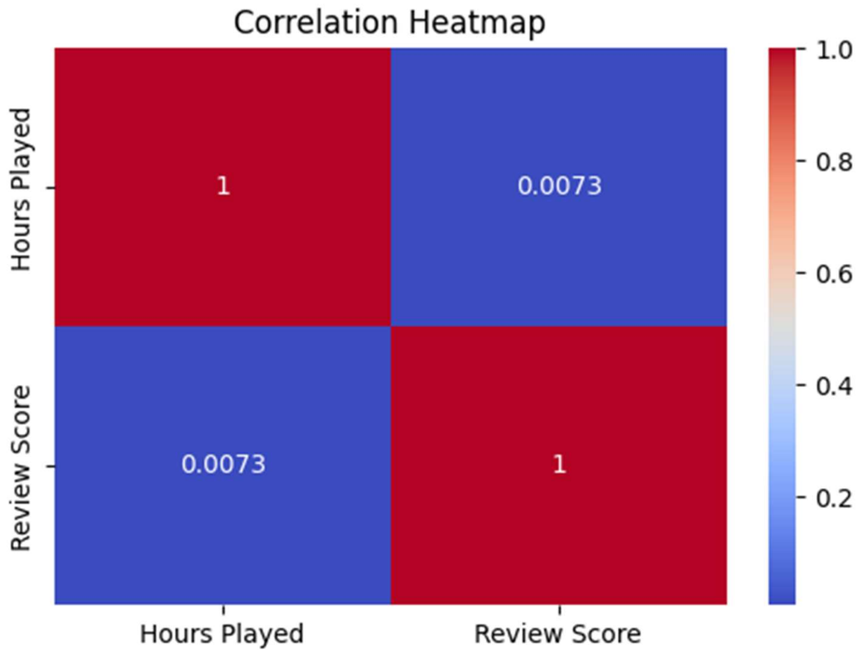
- Dataset screenshot (first 5–10 rows)
- Box plot charts
- Scatter plot with regression line
- Correlation heatmap
- Calculation snapshots (optional if formulas are shown)

Dataset Screenshot

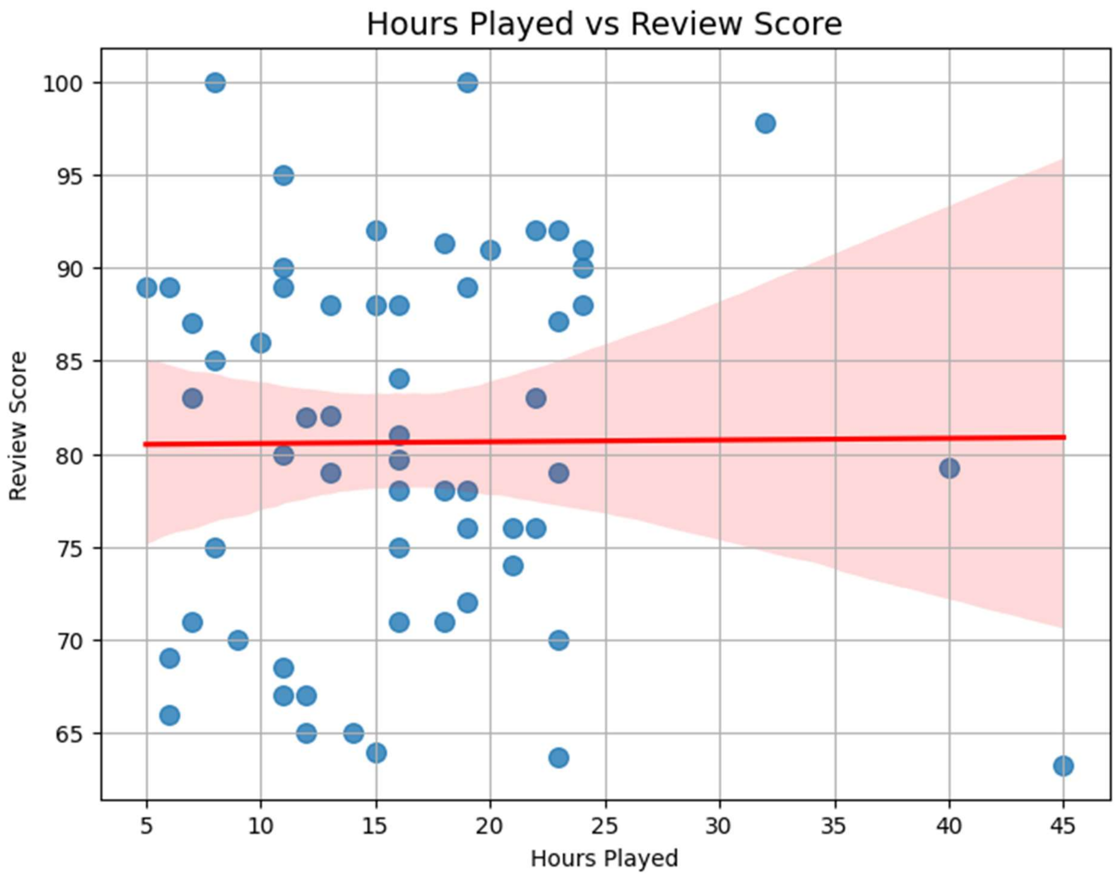
```
df.head(10)
```

	Player ID	Hours Played	Review Score
0	Player_1	11	67.0
1	Player_2	24	91.0
2	Player_3	19	100.0
3	Player_4	15	92.0
4	Player_5	12	82.0
5	Player_6	11	89.0
6	Player_7	23	70.0
7	Player_8	15	64.0
8	Player_9	15	88.0
9	Player_10	8	75.0

Correlation Heatmap



Scatter Plot with Regression Line



End of Report